

Signed impulse estimates and global smooth solutions for finite multispecies relativistic Vlasov–Maxwell

Alec Kriebel

ORCID: 0009-0001-9320-500X

October 6, 2026

Abstract

We extend the signed-impulse proof of the one-species three-dimensional relativistic Vlasov–Maxwell theorem in OpenAI’s family 362 manuscript to a fixed finite collection of particle species. Mass normalization gives a common velocity law and constant species-dependent accelerations. The entire source–receiver cancellation has coefficient $e_a e_b / m_a$; the source acceleration cancels before its sign is discarded. A positive mass-weighted energy controls a disjoint union of species label spaces, and the angular occupation and selected receiver coefficient use one simultaneous momentum bootstrap. We prove the resulting global classical theorem for nonnegative smooth compact phase data and compatible $C_b^\infty \cap L^2$ Maxwell fields, without smallness, symmetry or neutrality. The universal geometric identity is inherited; the new estimates keep all source and receiver coefficients explicit and close the shared-field bootstrap.

1 Problem, attribution and exact scope

We use rationalized units with light speed and vacuum permittivity one. For fixed $N \geq 1$, masses $m_a > 0$ and charges $e_a \in \mathbb{R}$, physical number densities $F_a(t, x, p)$ have velocities $p/(m_a^2 + |p|^2)^{1/2}$ and momentum accelerations $e_a(E + u_a \times B)$. Put

$$\begin{aligned} v &= p/m_a, & f_a(t, x, v) &= m_a^3 F_a(t, x, m_a v), \\ q(v) &= \sqrt{1 + |v|^2}, & u(v) &= v/q(v), & \lambda_a &= e_a/m_a. \end{aligned} \tag{1.1}$$

Thus $f_a dv = F_a dp$; f_a remains a number density. The equations are

$$\begin{aligned} \partial_t f_a + u \cdot \nabla_x f_a + \lambda_a (E + u \times B) \cdot \nabla_v f_a &= 0, & 1 \leq a \leq N, \\ \partial_t E - \operatorname{curl} B &= -j, & \partial_t B + \operatorname{curl} E &= 0, \\ \operatorname{div} E &= \rho, & \operatorname{div} B &= 0, \\ \rho &= \sum_a e_a \int f_a dv, & j &= \sum_a e_a \int u f_a dv. \end{aligned} \tag{1.2}$$

There is no background charge. A classical solution has C^1 densities and fields, solves these equations pointwise, has fields continuous into L_x^2 , and has compact particle phase support on each compact time interval.

Theorem 1.1 (Finite-species global theorem). *For every fixed finite species list as above, initial data*

$$0 \leq f_{a0} \in C_c^\infty(\mathbb{R}^6), \quad E_0, B_0 \in C_b^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L^2, \quad \operatorname{div} E_0 = \sum_a e_a \int f_{a0} dv, \quad \operatorname{div} B_0 = 0 \tag{1.3}$$

give a unique global classical solution of (1.2). It is smooth on each finite time interval, its particle phase support is compact there, and both constraints persist. On a finite maximal lifespan, a bound on $\max_a \sup_{(x,v) \in \text{supp } f_a(t)} |v|$ is a continuation criterion.

C_b^∞ means every spatial derivative is bounded. The field assumption does not require all derivatives to be square integrable. It permits a nonzero net charge and its Coulomb tail. Finite field energy supplies cone and spatial budgets, while bounded initial derivatives control homogeneous terms. Requiring all derivatives bounded is a convenient smooth-persistence class; no sharp minimal-regularity assertion is made. Constants below may depend on the data, the fixed horizon and all fixed masses and charges; no massless or infinite species limit is asserted. The first new special case is $N = 2$, $m_1 = m_2 = 1$, $e_1 = 1, e_2 = -1$. Noncompact momentum tails, collisions and curved spacetime are outside the theorem.

The underlying one-species unrestricted large-data theorem and the angular selection mechanism are due to the OpenAI manuscript [1], supplied as part of its October 2026 public repository release, with manuscript date September 23, 2026. We use its exact pinned version `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; the date printed on a manuscript is not evidence of an earlier public disclosure. This note's contribution is the signed cross-species estimate and its coupled closure, not an independent solution of the one-species problem. Glassey–Schaeffer [4] already use mass-normalized momentum and signed weighted source sums. Their large-data nearly-neutral theorem requires small signed phase-space density in C^1 and small compactly supported C^2 fields, rather than arbitrary signed data. Momentum normalization, multispecies formulation and bounded-momentum continuation are classical; see Glassey [2] and Luk–Strain [3].

If all charged species share a nonzero ratio $\lambda_a = \lambda$, set $g = \lambda^2 \sum_{e_a \neq 0} m_a f_a$ and $(\hat{E}, \hat{B}) = \lambda(E, B)$. Then $g \geq 0$ solves the unit-coefficient one-species system with these fields; its sources are the transformed Maxwell sources. Charged species can then be transported in that global field, and neutral species are free. This special case is already an immediate consequence of the inherited one-species theorem and known normalization. The extension requiring the estimates below has distinct charge-to-mass ratios, in particular the two unit-mass charges $+1, -1$.

Recent multispecies work of Mattingly–Pankavich–Ben-Artzi [7] sharpens decay and scattering for small globally neutral data. Its Theorem 1.1 recalls the earlier Glassey–Strauss small-data existence result; Theorems 1.5–1.6 concern scattering and decay. Those hypotheses do not duplicate the arbitrary-data, nonneutral theorem considered here.

2 Positive energy and separate label spaces

For species a , characteristics satisfy

$$Y'_a = u(V_a), \quad V'_a = \lambda_a(E + u(V_a) \times B). \quad (2.1)$$

The phase vector field has zero divergence, since $u = \nabla_v q$ and $\text{div}_v(u \times B) = 0$. Hence every flow preserves volume and transports f_{a0} , preserving positivity, particle number and its L^∞ norm. Integrating transport gives $\rho_t + \text{div } j = 0$ and propagates both Gauss constraints. Speeds below one give $\text{supp}_x f_a(t) \subset B(0, R_0 + t)$ for a common initial radius R_0 .

Define

$$\mathcal{U} = \frac{1}{2}(|E|^2 + |B|^2) + \sum_a m_a \int q f_a dv, \quad \mathcal{S} = E \times B + \sum_a m_a \int q u f_a dv. \quad (2.2)$$

Multiplying species transport by $m_a q$ gives work $m_a \lambda_a E \cdot \int u f_a dv = e_a E \cdot \int u f_a dv$. Its sum cancels the Maxwell work, including every charge sign. Thus $\partial_t \mathcal{U} + \text{div } \mathcal{S} = 0$, $|\mathcal{S}| \leq \mathcal{U}$, and

total energy $\mathcal{H}_0 = \int \mathcal{U}(0, x) dx$ is conserved. Spatial cutoffs with derivative $O(R^{-1})$ justify this identity because fields are $C_t L^2$ and particle support is compact on the prefix considered.

On a backward cone $y = x - (t - s)n$, put $r = t - s$ and $d = 1 - n \cdot u(v) > 0$. Its exact flux density is

$$\mathcal{U} - n \cdot \mathcal{S} = \frac{1}{2}(|E + n \times B|^2 + |n \cdot B|^2) + \sum_b m_b \int q df_b dv.$$

Integrating the shrinking-ball conservation law bounds its cone integral by $\mathcal{H}_0$. In particular, for $G = |E + n \times B| + |n \cdot B|$,

$$\int r^2 G^2 dn ds \leq 4\mathcal{H}_0, \quad \int r^2 \int q df_b dv dn ds \leq \mathcal{H}_0/m_b, \quad \int q f_b dx dv \leq \mathcal{H}_0/m_b. \quad (2.3)$$

All are positive bounds. No signed density is substituted for energy.

The positive aggregate $f_* := \sum_b f_b$ has $\|f_*\|_\infty \leq \sum_b \|f_{b0}\|_\infty$ and $\int q f_* \leq \mathcal{H}_0 \sum_b m_b^{-1}$, with the analogous cone bound. It generally solves no scalar Vlasov equation. Its label space is the *disjoint union* of the species label spaces, each with its own flow. Every labelwise calculation below uses the appropriate component before summing. This distinction is essential for opposite charges.

3 The signed source–receiver identity

Fix a supported receiver of species a , denoted $X(t), V_X(t)$, with $a(t) = u(V_X(t))$, $q_X = q(V_X)$ and $K_a = V'_X$. Use $Y_b(s, z), V_b(s, z)$ for a source label of species b . On its retarded branch,

$$t = s + r, \quad rn = X(t) - Y_b(s, z), \quad d = 1 - n \cdot u_b, \quad D = 1 - n \cdot a, \quad \eta = 1 - a \cdot u_b, \quad |n| = 1. \quad (3.1)$$

The symbol η is geometric, not a charge. Both d, D are positive. The retarded inverse flow has Jacobian $dz = d dy dv$: its derivative is the volume-preserving fixed-time inverse flow times $\text{Id} - (u_b, V'_b) \otimes (n, 0)$, whose determinant is d . Differentiating the retarded relation gives

$$\begin{aligned} \frac{dt}{ds} &= \frac{d}{D}, & ds f_{b0}(z) dz &= D dt f_b(s, y, v) dy dv, \\ r' &= d/D - 1, & rn' &= N_0 = (d/D)(a - n) + n - u_b. \end{aligned} \quad (3.2)$$

Strict timelikeness makes each noncolliding branch unique. The set of labels colliding with a fixed receiver at some equal time is null: a time mesh of size ζ covers such labels by $O(\zeta^{-1})$ spatial balls of volume $O(\zeta^3)$, and a bounded momentum set has finite volume. Volume preservation and bounded initial density give mass $O(\zeta^2)$. This argument applies separately to every source species.

Linearity of the wave potentials in Maxwell sources yields the source- b kernel, with $b_b = du_b/ds$,

$$\mathcal{K}_b = \frac{n - u_b}{r^2 q_b^2 d^3} + \frac{(n - u_b)(n \cdot b_b) - db_b}{r d^3}, \quad b_b = \frac{\lambda_b}{q_b} (\text{Id} - u_b \otimes u_b)(E + u_b \times B). \quad (3.3)$$

Its magnetic kernel is $n \times \mathcal{K}_b$ and its source multiplier is $e_b/(4\pi)$. The receiver force multiplier is consequently

$$c_{ab} = \lambda_a e_b = \frac{e_a e_b}{m_a}, \quad (3.4)$$

not $\lambda_a \lambda_b$. The latter normalization would omit m_b . Initial-cone and homogeneous-data terms are bounded by a datum/horizon constant, multiplied by the fixed coefficients. This follows from bounded initial momenta and Kirchhoff's estimate using bounded initial derivatives.

Set $Q = D\text{Id} + n \otimes a$, $H = Q/D$, $g = (u_b - n)/d$, and $k_0 = u_b - \eta n/D$. Then $Hg = k_0/d$. The complete branch-measure interior pair force, after removing $c_{ab}/(4\pi)$, is

$$\mathcal{K} = -\frac{H}{r} \partial_s^{(u_b)} g - \frac{Hg}{r^2 q_b^2 d}.$$

The second term is the transport term and must be retained.

Lemma 3.1 (Universal pair cancellation). *For arbitrary differentiable timelike source and receiver curves with the velocity law (1.1),*

$$\mathcal{K} = -\left(\frac{k_0}{rd}\right)' + \frac{\eta}{r^2 D^2} \left(\frac{n}{q_X^2 D} - a\right) + \frac{n(a_t \cdot k_0)}{r D^2}. \quad (3.5)$$

The identity requires no assumption on the signs or sizes of accelerations. It therefore holds after multiplication by every signed c_{ab} .

Proof. The product rule separates differentiation through u_b, n, r, a . For any variation A of a , $\partial_a k_0[A] = n(A \cdot k_0)/D$; since $a' = a_t d/D$, this gives the last term. The remaining geometric identities are

$$N_0 \cdot u_b - dr' - q_b^{-2} = -\eta d/D, \quad rd(k_0)'_n = -\frac{\eta d}{D} \left(N_0 + \frac{n(N_0 \cdot a)}{D}\right),$$

$$N_0 + \frac{n(N_0 \cdot a)}{D} + k_0 = \frac{d}{D} \left(a - \frac{n}{q_X^2 D}\right).$$

They follow by expanding (3.1), using $|u_b|^2 = 1 - q_b^{-2}$ and $|a|^2 = 1 - q_X^{-2}$. Substitution into the differentiated primitive gives (3.5). These identities are also checked by the denominator-cleared universal polynomial certificate distributed with this note. The proof, rather than the supplementary finite rational samples, establishes all quantifiers. $\square$

After conversion to $dt dy dv$, the nondifferentiated residuals are

$$\mathcal{R}_{ab} = c_{ab} \frac{f_b \eta}{r^2 D} \left(\frac{n}{q_X^2 D} - a\right), \quad \mathcal{A}_{ab} = c_{ab} \frac{f_b n(a_t \cdot k_0)}{r D}, \quad a_t = \frac{\text{Id} - a \otimes a}{q_X} K_a. \quad (3.6)$$

Here K_a is the *full* acceleration from the common field. No additional factor λ_a belongs in this last formula.

4 Uniform occupation and coefficient transfer

We spell out why the analytic estimates of [1, Sections 4–8] apply to this collection, rather than invoking its one-species theorem on separately forced populations.

Fix a horizon T_0 and a compact classical prefix $[0, t_*]$ with $t_* \leq T_0$. Choose a sufficiently large dyadic P such that

$$P \geq 1024 \max_{b, z \in \text{supp } f_{b0}, t \leq t_*} q(V_b(t, z)).$$

Empty label components are omitted. Write

$$L = \log(2 + P), \quad S(w) = P^2 L/w.$$

The simultaneous bootstrap assumes, for every species and supported label,

$$|V_a(t_2) - V_a(t_1)| \leq MP\sqrt{I} + AIS(w), \quad I = t_2 - t_1, \quad w/8 \leq q(V_a(t)) \leq 8w \quad (t \in [t_1, t_2]). \quad (4.1)$$

One chooses M, A only after establishing constants independent of them.

Use the exact dyadic partitions of [1, Section 4]: $q_b \asymp p$, $q_X \asymp w$, $d \asymp \theta^2$, $D \asymp \phi^2$, $\eta \asymp \sigma^2$, $r \asymp h$ after normalizing the radius by $\max(1, T_0)$. Source representatives are $p = 64 \cdot 2^k$; the fixed comparisons imply $64 \leq p \leq P$, $\theta \geq 1/p$, $\phi \gtrsim 1/w$, $\sigma \gtrsim 1/m$, $m = \min(p, w)$. For $\theta < \kappa\phi$ the source/receiver separation scale is $\sigma \asymp \phi$ and its partition is omitted; otherwise retain σ and put $\nu = \min(\theta, \sigma)$, $k_1 = 1 + \phi/\sigma$, $\omega = \sigma$. In the small sector put $\nu = \theta$, $k_1 = 1$, $\omega = \phi$. The fixed κ is small enough for all enlarged overlapping supports.

For a receiver interval of length at most $I \leq (w/P)^2$, define the aggregate hit measure $M = \sum_b \int r^2 DN_b dt ds dn$, where N_b is the bin-restricted velocity integral of f_b . Each species has $N_b \leq C_b p^3 \nu^2$, by its conserved L^∞ norm and the velocity cap. At fixed source time the spatial map $(t, n) \mapsto X(t) - (t - s)n$ is injective with Jacobian $r^2 D$. By (3.2),

$$M = \sum_b \int f_{b0}(z) dz \int_{\text{bin hits}} ds, \quad M \leq C \max(I, h)/p, \quad M \leq CI\phi^2/\theta^2. \quad (4.2)$$

The energy bounds in (2.3) prove the first estimate; branch time change and the finite sum of species particle numbers prove the second.

For a sufficiently large fixed C_1 , set

$$\ell^{-1/2} = C_1 \frac{P}{m} \left(\frac{M}{\omega} + \sqrt{\frac{AL}{\omega}} \right). \quad (4.3)$$

The first-exit argument using (4.1) bounds source and receiver increments on cells of length $O(\ell)$ by $\varepsilon m \omega$. The velocity derivative has norm at most $1/q$, giving velocity changes $\varepsilon \omega$. This step uses each species' own signed bootstrap; there is no equality of its acceleration with another species' acceleration. On actual bins $\omega > C_0/m$, projected relative velocity is bounded below by $c\omega$ on a cell. The same-time separation at a hit is $O(h(\phi + \omega))$, so monotone crossing gives duration Chk_1 per label. The label mass at a common cell time is at most C/p after summing the species energy bounds. Consequently

$$M \leq C(hk_1/p) \max(1, I/\ell, h/\ell). \quad (4.4)$$

For the small sector, average the cell mass using the two indicators $q \asymp p$, $|u - a(s)| \asymp \phi$ and $q \asymp p$, $\sqrt{(1 - n_{\text{ret}} \cdot a(t_{\text{ret}}))/2} \asymp \phi$ over one enlarged source-time interval of length $O(I)$. Define $\mathcal{E}_{p\phi}$ with density qf_* and normalization $1/I$. Both indicator families have bounded overlap and unique retarded receiver time, so $\sum_{p,\phi} \mathcal{E}_{p\phi} \leq C$. Full stable cells can be averaged; the at most two clipped cells use their direct mass bound. When $\ell < h \leq I$, the spatial hit map and the second indicator give $M \leq CI\mathcal{E}_{p\phi}/p$ directly. These arguments yield

$$M \leq C(hk_1/p)R^2, \quad R = 1 + \sqrt{I/\ell}\sqrt{\mathcal{E}}, \quad \mathcal{E} = \mathcal{E}_{p\phi} \text{ in the small sector, } \mathcal{E} = 1 \text{ otherwise,} \quad (4.5)$$

except the deficit bins $\omega \leq C_0/m$ with $h \leq \min(I, \ell)$. On that exception $\ell \leq CP^{-2}$ and the direct near bounds suffice. Every cell and indicator proof operates on each component of the disjoint label space, then adds positive masses. Thus the scalar transport equation for f_* is never assumed.

4.1 Direct bounds, derivatives and uniform constants

The source acceleration satisfies

$$|b_b| \leq C|\lambda_b|q_b^{-1}(G + \theta^2|B|), \quad |n \cdot b_b| \leq C|\lambda_b|\theta q_b^{-1}(G + \theta^2|B|), \quad |q'_b| \leq C|\lambda_b|(G + \theta|B|).$$

For every pair the transport majorants are $C|c_{ab}|f_b/(r^2p^2\theta^2)$ and $C|c_{ab}|f_b\phi/(r^2p^2\theta^3)$; the good-field and magnetic majorants are

$$C|c_{ab}\lambda_b|\frac{f_b\phi G}{rp\theta^2}, \quad C|c_{ab}\lambda_b|\frac{f_b\phi|B|}{rp}. \quad (4.6)$$

Perpendicular receiver projection supplies an additional factor ϕ to the acceleration-kernel majorants, because the kernel numerator is tangent to n . Taking the finite maximum of $1, |c_{ab}|, |c_{ab}\lambda_b|$ bounds all these by the same positive majorants with density f_* , up to a fixed constant.

Cone G^2 budgets and spatial B^2 budgets are shared over the angular bins, as in [1, Section 5]. Cauchy–Schwarz, the density cap and (4.5) give the following representative near/far bounds in the small sector, suppressing their fixed constants and the square roots of bounded field budgets:

$$\begin{aligned} T_\perp : \quad & \min(Ihp\phi^3/\theta, M/(h^2\phi p^2\theta^3)), \quad S_g : \quad \min(I\sqrt{h}p^2\phi^2, \sqrt{I}\sqrt{pM}/(\theta h)), \\ S_b : \quad & \min(\sqrt{I(I+h)}\sqrt{h}p^2\theta^2\phi, \sqrt{I+h}\sqrt{pM}\theta/(\phi h)). \end{aligned}$$

The remaining sector replaces the relevant θ^2 density factor by ν^2 , and retains k_1 . These are exactly the nonnegative inputs to the radial balances and angular sums of [1, Section 5]. For example $\sum_h \min(Ah^\alpha, Bh^{-\beta}) \leq C_{\alpha,\beta} A^{\beta/(\alpha+\beta)} B^{\alpha/(\alpha+\beta)}$. The finite species constants multiply A, B outside these balances. They do not alter any exponent, shared-budget overlap or bootstrap parameter. The resulting baseline and absolute bounds are

$$\begin{aligned} \text{baseline cost} &\leq C\mathcal{B}(I, w), \quad \int_J |K_a| dt \leq C\sqrt{w}\mathcal{B}(I, w), \\ \mathcal{B}(I, w) &= P\sqrt{I} + (M + \sqrt{A})IS(w). \end{aligned} \quad (4.7)$$

Here baseline means transport, magnetic remainder and the remaining-sector good-field cost, as precisely enumerated in that source proposition. The exceptional deficit near costs have ratio $O(L^3/P)$ to $IS(w)$.

The signed primitive creates source derivatives with coefficient $|c_{ab}\lambda_b|$ and exactly the majorants (4.6); geometric cutoff derivatives have majorant $C|c_{ab}|f_b\phi/(r^2\theta)$. The central residual is bounded by $C|c_{ab}|f_b/r^2$. The receiver residual and receiver-cutoff derivatives have majorant

$$C|c_{ab}|\frac{f_b\theta|K_a|}{wr\phi^2}. \quad (4.8)$$

Projection adds ϕ to these bounds. For the moving projector $\Pi = w(\text{Id} - \xi \otimes \xi)/|V_X|$, $\xi = V_X/|V_X|$, its derivative has the projected receiver size. A Lipschitz weight derivative contributes $C|c_{ab}||\psi_t|f_b\theta/(r\phi)$.

Choose representative selections fixed for the receiver interval and common to every source component. Derivatives of their selected partition sum equal the negative sum of unselected derivatives, with common physical multiplier factored out. Supports meeting have representatives within factor four. Thus source and geometric derivative costs can be assigned to unselected neighbors as in [1, Section 6]; no different kernels or charges are canceled against each other. Integration by parts is valid componentwise. Initial-cone terms have total cost $C\int|\psi|$, since $Dk_0 = Du_b - \eta n$ is bounded and each initial d is bounded below. A radial tip cutoff has error $O_{\text{prefix}}(\gamma)$ and can be removed on the fixed prefix. Such justification constants do not enter the uniform estimates.

4.2 Direction occupation and selected coefficients

Apply the projected identity with selection $\theta < \kappa\phi$, $p\theta\sqrt{h} \leq 1$ to the successive first exits of direction of a receiver of *any* species. For $w \geq P^{3/5}$, $w^{-3/5} \leq \delta \leq w^{-1/10}$ and a time interval of length at most $(w/P)^2$, the direction count proof of [1, Section 7] uses only (4.1), (4.5), (4.7) and the projected majorants above. The total receiver/projector coefficient is CL^2/w ; the time-weight cost is CL^2 per exit. Bulk terms are integrated over disjoint intervals, so they share one length bound rather than gaining an exit count. The estimate is

$$\frac{1}{2}N_c w \delta \leq C(M, A)wL^6 + CL^2N_c.$$

Absorb the last term using $w\delta \geq P^{6/25}$ to obtain $N_c + 1 \leq C(M, A)L^6/\delta$. This uses no improved occupation input.

Separate source and receiver direction partitions then give, in the window

$$p, w \geq P^{7/10}, \quad j^{-49/100} \leq \phi \leq j^{-4/25} \quad (j = p, w), \quad (4.9)$$

$$M \leq C(h/p)\phi^{-29/25} \max(1, I/I_m^0, h/I_m^0), \quad I_m^0 = (m/P)^2. \quad (4.10)$$

Indeed, with stable energy cells of length $I_m^0/[C_2(M^2 + AL)]$, bad hits crossing a receiver direction boundary have source duration $O(h)$ per boundary. Good hits have same-time separation $O(h\phi)$ and monotone relative velocity $\Omega(\phi)$ on the common refinement of source and receiver direction cells. The refinement count is their sum, so duration is $C(M, A)hL^6/\phi$ per label. Summing positive species masses and cells gives the pre-absorption cost $C(M, A)(h/p)\phi^{-1}L^7 \max(1, I/I_m^0, h/I_m^0)$. The window gives $\phi^{-4/25} \geq P^{56/3125}$, absorbing the fixed $C(M, A)L^7$ for a sufficiently large threshold. The final C in (4.10) is independent of M, A . This is the required order of dependency, rather than a circular use of improved occupation to count directions.

Set $\epsilon = 10^{-5}$, $\beta = 29/25$ and $W = (P\theta)^{-\epsilon}\phi^\epsilon h^\epsilon$. Let M_* be the minimum of the *pure* entries

$$\frac{\max(I, h)}{p}, \quad \frac{h}{p} \max(1, I/\ell, h/\ell) \quad (\phi > C_0/m), \quad \frac{h}{p} \phi^{-\beta} \max(1, I/I_m^0, h/I_m^0) \text{ when (4.9) holds.}$$

Then $M \leq CM_*$, with the constant outside the entries. Select a small-sector representative if

$$\min \left(I\sqrt{h}p^2\phi^2, \frac{\sqrt{I}\sqrt{pM_*}}{\theta h} \right) > P\sqrt{I}W. \quad (4.11)$$

Unselected good-field costs sum to $CP\sqrt{I}$ because $\sum W \leq C$. At fixed receiver time, (4.8), the density cap and the positive cone budget give aggregate coefficient at most CU , where

$$U = w^{-1} \min\{h^2(p\theta)^3, (p\theta h\phi^2)^{-1}\}. \quad (4.12)$$

Both estimates are available with fixed species-dependent constants. Taking their maximum bounds the minimum by CU . The exact representative argument of [1, Section 8] gives

$$U \leq w^{-1/2}W \quad \text{on selected representatives for } P \geq P_*(M, A). \quad (4.13)$$

To check its decisive compatibility, write $p = P^{1-\alpha}$, $w = P^z$, $c = 1 - z$, $p\theta = P^b$, $\phi = P^{-y}$, $h = P^{-H}$ and $\Delta = \epsilon(\alpha + b + y + H)$. Near selection gives $2\alpha + H/2 + 2y < z + \Delta$. Energy far selection and unsafety force $\Delta < 4z/99995 < .001z$, $H > .496z + 3(\alpha + c)$ and actual separation $m\phi > P^3$. Stable-cell far selection then forces the window (4.9). Improved occupation and unsafety imply

$$\beta y > H/3 + z/3 + 2\min(\alpha, c) - 8\Delta/3, \quad \beta y < 29z/50 - 29H/100 + (29/50)\Delta.$$

Consequently $H/z < 37487/93500 < 62/125$, contradicting $H/z > .496$. The intervening window inequalities and exact rational margin are checked by the accompanying selection certificate and reproduced in the audit supplement; no species coefficient occurs inside these pure inequalities.

Summing (4.13) bounds the coefficient of the *full* $|K_a|$ by $Cw^{-1/2}$, uniformly over all receivers. It cancels the $\sqrt{w}$ loss in (4.7). The spatial, central and source derivative balances and endpoint-weight estimates of [1, Section 8] use exactly the transferred majorants listed above. Hence, for an endpoint-zero weight with ramps of length $\eta_0 I$,

$$\left| \int_J \psi K_a dt \right| \leq C_{\eta_0} [P\sqrt{I} + (M + \sqrt{A})IS(w)]. \quad (4.14)$$

The constant depends on the fixed species data but not on M, A . This proves parameter robustness of the analytic chain: positivity enters only the aggregate majorants and energy, while every signed kernel identity was first taken pairwise with its full fixed coefficient.

5 Coupled closure and exclusion of finite-time blowup

Let $\mathcal{B}_w(I) = MP\sqrt{I} + AIS(w)$. If it is at least $32w$, the endpoint energy range directly gives $|\Delta V_a| \leq 16w \leq \mathcal{B}_w(I)/2$. Otherwise $M \geq 32$ implies $I \leq (w/P)^2$. Restore the two edge ramps in (4.14) by integrating the removed weight against $V_a(t) - V_a(t_1)$ on the left and $V_a(t) - V_a(t_2)$ on the right. Each monotone ramp has derivative norm one, so the simultaneous signed bootstrap, rather than an absolute force bound, gives total cost $2\mathcal{B}_w(\eta_0 I)$. Thus

$$|\Delta V_a| \leq (C_{\eta_0} + 2M\sqrt{\eta_0})P\sqrt{I} + [C_{\eta_0}(M + \sqrt{A}) + 2A\eta_0]IS(w).$$

Choose $2\sqrt{\eta_0}, 2\eta_0 \leq 1/8$, then $M \geq \max(32, 8C_{\eta_0})$, then $A \geq \max(1, 16C_{\eta_0}M, 256C_{\eta_0}^2)$, and finally P_0 large enough for all preceding estimates. This gives a strict half-bootstrap for *every* species with one set of constants.

The set of times up to which all eligible intervals and all species satisfy (4.1) is closed by truncation and continuity. On a fixed compact classical prefix, the finite union of supported label sets has a finite acceleration bound C_* . Every interval of sufficiently small length obeys $C_*I \leq MP\sqrt{I}$. The strict improvement on a longer interval's prefix and $C_*\varepsilon$ on its suffix prove relative right openness, uniformly over the finite species list. This temporary C_* affects only the openness argument; it never enters M, A, P_0 . Therefore the signed increment estimate is unconditional on every compact classical prefix, with constants independent of that prefix's endpoint.

If all particle data vanish, smooth vacuum Maxwell evolution is global. Otherwise suppose a maximal solution had finite lifespan and unbounded joint momentum. Let $Q(t) = \max_{a,z \in \text{supp } f_{a0}} q(V_a(t, z))$ and t_n its first hit of 2^n for large n . Choose a maximizing species and label at t_n , and s_n its last hit of 2^{n-1} . Then $s_n \geq t_{n-1}$ even if the maximizer changes species, and the chosen label remains in $[2^{n-1}, 2^n]$ on $[s_n, t_n]$. Apply the signed estimate with $P = 1024 \cdot 2^n$, $w = 2^n$, $\Delta_n = t_n - s_n$. Since q is one-Lipschitz,

$$2^{n-1} \leq 1024M2^n\sqrt{\Delta_n} + 1024^2A2^n\Delta_n \log(2 + 1024 \cdot 2^n).$$

One term must exceed 2^{n-2} , so $t_n - t_{n-1} \geq \Delta_n \geq c/\log(2 + 1024 \cdot 2^n)$. The sum diverges. Finite-time momentum blowup is impossible.

6 Local theory, uniqueness and nonneutral continuation

For Sobolev fields we use the classical bounded-momentum criterion of Glassey–Strauss [5], in the precise formulation of Luk–Strain [3, Theorem 1.1, Footnote 1, Remark 1.2]: nonnegative

compact phase data in H^5 , fields in H^5 , Gauss constraints, and a uniform finite momentum radius imply continuation. Their remark explicitly extends the method to finitely many species. Glassey's book [2, Chapter 5] formulates the masses and signs and explains the species-specific streaming operators and separate iterate differences. Its printed theorem includes neutrality; we do not silently remove that hypothesis from its statement.

The finite-species Sobolev adaptation uses the same bounded-radius velocity law after (1.1). Source field representations acquire $e_b \lambda_b$ and receiver force equations acquire λ_a ; absolute estimates replace them by fixed constants and sum finitely many species. At bounded radius R , all cone divisors obey $1 - n \cdot u \geq 1 - R/\sqrt{1+R^2} > 0$. The division-lemma and differentiated kernel estimates in [6, 3] therefore have the same bounded constants and logarithmic derivative inequality, with $\mathcal{N}(t) = \max_{s \leq t} \sum_a \|\nabla_{x,v} f_a(s)\|_\infty$:

$$\|(E, B)(t)\|_{W^{1,\infty}} \leq C[1 + \log(2 + \mathcal{N}(t))], \quad \mathcal{N}(t) \leq \mathcal{N}(0) + C \int_0^t [1 + \log(2 + \mathcal{N}(s))] \mathcal{N}(s) ds.$$

The Osgood integral diverges, controlling derivatives; higher Sobolev norms then satisfy the usual tame transport–Maxwell Gronwall estimate. Local H^6 iteration uses $\|j\|_{H_x^k} \leq |B_R|^{1/2} \sum_a |e_a| \|f_a\|_{H^k}$ and the divergence-free phase transport. It gives a common local lifespan, positivity and smooth persistence for smooth data. The constants depend on R ; these continuation estimates do not replace the a priori signed proof.

To reach (1.3), fix any finite horizon T and choose $L_0 > R_0 + 2T$. The signed smooth compact initial density ρ_0 has Coulomb field $E_C = -\nabla(-\Delta)^{-1}\rho_0$. It is smooth and belongs to every H^k in three dimensions: its Fourier transform is $-i\xi\hat{\rho}_0/|\xi|^2$, whose $O(|\xi|^{-1})$ behavior is square integrable near zero, and $\hat{\rho}_0$ decays rapidly at infinity. This argument permits $\int \rho_0 \neq 0$.

Both $G_E = E_0 - E_C$ and $G_B = B_0$ are smooth divergence-free fields. For either such G , the smooth potential

$$A_G(x) = \left(\int_0^1 tG(tx) dt \right) \times x$$

satisfies $\text{curl } A_G = G$, as follows by differentiating $t^2G(tx)$ and using $\text{div } G = 0$. Let a compact cutoff χ equal one on $B(0, L_0)$. Set $\tilde{E}_0 = E_C + \text{curl}(\chi A_{G_E})$ and $\tilde{B}_0 = \text{curl}(\chi A_{G_B})$. These fields lie in every H^k , preserve both constraints and coincide with the originals on $B(0, L_0)$.

The difference is divergence-free smooth L^2 data for vacuum Maxwell. Its global vacuum evolution W is $C_t L^2$, smooth with bounded derivatives, and vanishes on $|x| < L_0 - t$ by finite propagation. For $t \leq T$ it vanishes throughout every particle region $|x| \leq R_0 + t$. Adding W to a solution for modified data, or subtracting it from an original-data solution, preserves all species transport equations on their supports, Maxwell equations and constraints, without changing momenta. Thus bounded-momentum continuation and smooth local theory transfer to (1.3) on any prescribed horizon.

Uniqueness holds in the full classical class, not only the Sobolev construction. On a common compact interval choose a common momentum radius R and subtract two solutions. Transport difference energies are bounded by $C \sum_a \|\delta f_a\|_2 \|\delta(E, B)\|_2$, since the other solution's first density derivatives are bounded on compact phase support. Maxwell's difference current satisfies $\|\delta j\|_2 \leq |B_R|^{1/2} \sum_a |e_a| \|\delta f_a\|_2$. Spatial cutoffs justify Maxwell energy without assuming L^2 field derivatives. Gronwall applied to $\sum_a \|\delta f_a\|_2^2 + \|\delta(E, B)\|_2^2$ gives equality. Together with the signed estimate and momentum-doubling argument this completes the proof of Theorem 1.1. The dependency ledger and independent adversarial audit records describe the validation performed.

7 Checks, limitations and disclosure

If a receiver has $e_a = 0$, its momentum is constant and all pair factors vanish. A source with $e_b = 0$ has free transport and no Maxwell contribution; its positive energy may be retained harmlessly. Identical mass/charge species can be combined by adding number densities. For a single species with $m = e = 1$ every displayed formula reduces to [1]. For two unit masses and charges $+1, -1$, same-sign pair coefficients are $+1$ and cross-sign coefficients -1 ; source accelerations are signed before exact cancellation. An extreme fixed mass ratio only enlarges m_b^{-1} , $|\lambda_b|$ and $|c_{ab}|$ and the normalized initial norms. There is no uniform estimate as a mass approaches zero. Neutrality is used nowhere in the positive energy, kernels, occupation or localization proof.

The assumptions of bounded smooth field derivatives provide the homogeneous field and vacuum regularity needed by this proof. Arbitrary smooth finite energy fields without those derivative bounds are not covered. Compact particle phase support is a hypothesis of the analytic chain. The note does not infer control of support from energy alone, or a scalar Vlasov equation for $\sum_b f_b$.

The source package contains exact source hashes, dependency and priority ledgers, scoped agent audits, and standard-library kernel and exponent certificates. Those certificates check algebra only. This multispecies theorem is not claimed to be Lean formalized; a source-library static scan or existence of comparator files is not formal certification. AI tools were used extensively in the research, drafting and verification. Automated adversarial reviews do not constitute conventional human peer review. This preprint has not undergone conventional human peer review or refereeing. The ordinary analytic proof is the basis for the theorem; no reproduced Lean certificate is claimed.

References

- [1] OpenAI. *Global classical solutions of the three-dimensional relativistic Vlasov–Maxwell system*. OpenAI Math Release preprint, manuscript dated September 23, 2026. Pinned repository input: `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, family 362. The manuscript-specific citation is supplied in that family’s README.
- [2] R. T. Glassey. *The Cauchy Problem in Kinetic Theory*. SIAM, 1996, Chapters 5–6. doi:10.1137/1.9781611971477.
- [3] J. Luk and R. M. Strain. A new continuation criterion for the relativistic Vlasov–Maxwell system. *Commun. Math. Phys.* 331 (2014), 1005–1027. arXiv:1406.0165v1; doi:10.1007/s00220-014-2108-8.
- [4] R. T. Glassey and J. Schaeffer. Global existence for the relativistic Vlasov–Maxwell system with nearly neutral initial data. *Commun. Math. Phys.* 119 (1988), 353–384. doi:10.1007/BF01218078.
- [5] R. T. Glassey and W. A. Strauss. Singularity formation in a collisionless plasma could occur only at high velocities. *Arch. Ration. Mech. Anal.* 92 (1986), 59–90. doi:10.1007/BF00250732.
- [6] F. Bouchut, F. Golse and C. Pallard. Classical solutions and the Glassey–Strauss theorem for the 3D Vlasov–Maxwell system. *Arch. Ration. Mech. Anal.* 170 (2003), 1–15. arXiv:math/0301175.

- [7] G. Mattingly, S. Pankavich and J. Ben-Artzi. Sharp time decay and scattering of small data solutions to the relativistic Vlasov–Maxwell system. arXiv:2609.29013v1, submitted September 24, 2026.